Timothy James R. Brannon

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EDUCATION

Clemson University — Clemson, SC
B.S. Computer Engineering. GPA: 3.26/4.00. — *Expected December 2026*

Greenville Technical College — Greenville, SC
A.S. in Computer Engineering. GPA: 3.66/4.00. — *December 2023*

SKILLS

Embedded: ESP32, Arduino, Embedded C/C++, ESP-NOW, I2S, Sensors, Interrupts
Programming: C, C++, MATLAB, Python, SQL, Java
Engineering: DC Motor Control, Feedback Control, Computer Vision, Signal Processing
Software: React Native, Expo, PostgreSQL, Supabase, Git, GitHub
Tools: MATLAB App Designer, SolidWorks, Arduino IDE

PROJECTS

IoT Smart Light Detection and Audio Alert System

- Developed a reliable two-node ESP32 system using light and door-state sensors to detect when a light may have been left on, with the architecture designed to scale to additional rooms and sensor types.
- Implemented ESP-NOW communication, embedded decision logic, and audio alerts using I2S, MAX98357A, and flash-based audio storage.

Automated Gameboard

Senior Design I

- Built software integrating DC motor control, feedback control, computer vision, game logic, and a MATLAB GUI into a unified automated system as part of a five-person team of 2 computer engineers and 3 electrical engineers.
- Implemented closed-loop motor control and computer vision to identify dice outcomes and automate game decisions; successfully demonstrated in a recorded working demo.

Throne Finder

Mobile Application

- Building a cross-platform mobile application for locating and rating public restrooms through an interactive map.
- Designing a PostgreSQL/Supabase backend with authentication, user data management, and row-level security.
- Managing development and version control using Git and GitHub.

EXPERIENCE

Car Porter — May 2025 – Present
Fairway Ford — *Greenville, SC*

- Coordinate vehicle movement, customer pickup/delivery, and communication between customers and service advisors.
- Manage vehicle transfers while maintaining safety and accuracy.

Team Member — August 2023 – April 2024
Chick-fil-A — *Taylors, SC*

- Executed service operations in a high-volume environment, prioritizing accuracy, teamwork, and rapid decision-making.

Sales Associate — August 2019 – May 2023
Mast General Store — *Greenville, SC*

- Trained new employees and developed organizational systems to improve efficiency and reduce errors.

AWARDS & HONORS

- President's List, Greenville Technical College
- Dean's List, Multiple Semesters
- Volunteer, Samaritan's Purse Tornado Cleanup
- Youth Ministry Intern, Rock Springs Baptist Church